Notes: Arellano, Bai, and Mihalache (QJE, 2026)

Monetary Policy and Sovereign Risk in Emerging Economies (NK-Default)

Contents

1 Big picture

- **Question.** How do monetary policy and sovereign default risk interact in emerging-market economies that use inflation-targeting interest rate rules?
- **Core gap in standard models.** Open-economy New Keynesian models are central to monetary policy analysis, but they typically abstract from sovereign risk. Sovereign-default models study fiscal borrowing and spreads, but they usually abstract from sticky prices and monetary policy.
- **Main contribution.** The paper integrates a small-open-economy New Keynesian model with long-term foreign-currency sovereign debt and endogenous default.
- **Key empirical motivation.** In emerging-market inflation-targeting countries, inflation, sovereign spreads, and domestic policy rates often rise together during temporary inflation events, while output falls.
- **Mechanism 1: default amplification.** A rise in default risk raises expected future marginal utility and expected future inflation. Forward-looking firms increase prices today, and households reduce current consumption. The result is higher current inflation and a higher monetary wedge.
- **Mechanism 2: monetary discipline.** Tight monetary policy raises the real costs of fiscal overborrowing. Because the government internalizes monetary distortions, higher nominal rates can reduce borrowing incentives, lower default risk, and reduce sovereign spreads.
- **Policy trade-off.** Strict inflation targeting eliminates price-setting distortions in the short run, but it may leave excessive sovereign borrowing and default risk. A monetary rule that reacts to default risk can improve welfare.
- **Quantitative result.** Default risk accounts for a large share of inflation volatility in the calibrated emerging-market economy. The baseline monetary rule reduces spreads relative to strict inflation targeting.
- **Welfare result.** Delivering the flexible-price allocation is not necessarily optimal for monetary policy when the sovereign can default. Some monetary distortions can be useful because they discipline fiscal borrowing.
- **Takeaway.** In emerging economies, monetary policy is not only about stabilizing inflation and output. It also affects sovereign risk, borrowing incentives, and spreads.

2 Data and measurement

2.1 Emerging-market inflation-targeting sample

- The empirical sample consists of eight emerging-market inflation targeters: Brazil, Chile, Colombia, Mexico, Peru, the Philippines, Poland, and South Africa.
- The sample period is 2004Q1–2019Q4, by which point the countries had adopted inflation targeting.
- The sample intentionally excludes the post-2020 global inflation episode from the baseline data analysis.
- The countries are chosen because they appear in the JP Morgan EMBI+ and have an operational inflation-targeting regime.

Interpretation: The data sample is designed to focus on countries where monetary policy resembles an interest-rate-based inflation-targeting regime but sovereign spreads remain economically important.

2.2 Macroeconomic variables

- **Inflation:** CPI four-quarter log difference.

- **Spreads:** EMBI+ spreads on foreign-currency sovereign bonds relative to comparable U.S. government bonds.
- **Domestic nominal rate:** central bank policy rate.
- **Output:** four-quarter log difference of real GDP.

Interpretation: Spreads are measured on foreign-currency sovereign debt, while policy rates are in domestic currency. This means the positive relation between policy rates and spreads is not mechanically driven by domestic-currency inflation or devaluation.

2.3 Stylized facts

- Mean inflation across countries averages about 3.9%.
- Mean sovereign spreads average about 2.0%.
- Inflation volatility relative to output volatility averages 0.8.
- Spread volatility relative to output volatility averages 0.4.
- Spreads are positively correlated with inflation, with an average correlation around 0.5.
- Spreads are positively correlated with domestic nominal policy rates, with an average correlation around 0.3.
- Spreads are negatively correlated with output, with an average correlation around -0.5.

Interpretation: These facts motivate a framework in which sovereign risk affects nominal outcomes and monetary policy reacts to inflation while also shaping default risk.

2.4 Temporary inflation events

- A country is classified as having a temporary inflation event when inflation is at least two country-specific standard deviations above its mean.
- Events are centered on the quarter of peak inflation.
- The paper studies average paths from eight quarters before to eight quarters after the peak.
- Inflation rises about 2.5 standard deviations at the peak, equal to roughly 4.5 percentage points.
- The domestic nominal rate rises about 1.2 standard deviations, equal to roughly 2.9 percentage points.
- Output falls about 2.5 standard deviations, equal to about a 5.7 percentage-point recession.
- Spreads rise about 2.5 standard deviations, equal to roughly 2.3 percentage points.

Interpretation: These events are short-lived but severe. Inflation, spreads, and policy rates all spike, and output contracts. The model is built to rationalize exactly this joint pattern.

3 Model primitives and equilibrium objects

3.1 Environment

- The economy is a small open economy with households, final-goods firms, intermediate-goods firms, a monetary authority, and a fiscal government.
- There are domestic final goods, domestic intermediate varieties, and foreign imported goods.

- Intermediate firms produce differentiated goods and set prices subject to Rotemberg price-adjustment costs.
- The government borrows internationally using long-term foreign-currency debt and may default.
- Monetary policy sets the domestic nominal interest rate according to an inflation-targeting rule.

Interpretation: The model joins two normally separate frameworks: a sticky-price monetary model and a sovereign-default model with endogenous borrowing and spreads.

3.2 Terms of trade and export demand

The terms of trade are

$$e_t = \frac{P_t^f}{P_t^d} = \frac{\varepsilon_t P_t^*}{P_t^d}, \tag{1}$$

where ε_t is the nominal exchange rate, domestic-currency units per foreign-currency unit. The foreign price level is normalized to one.

Export demand is

$$X_t = e_t^{\rho} \xi, \tag{2}$$

where ρ is the export demand elasticity and ξ is the level of foreign demand.

Interpretation: A depreciation raises the terms of trade and stimulates exports. Export demand matters because monetary tightening can appreciate the real exchange rate and reduce output.

3.3 Households

Households choose domestic consumption C_t, foreign-goods consumption C_t^f, labor N_t, and domestic bond holdings. Their flow utility is

$$u(C_t, C_t^f, N_t). \tag{3}$$

The key optimality conditions are

$$-\frac{u_{N,t}}{u_{C,t}} = w_t, \tag{4}$$

$$\frac{u_{C^f,t}}{u_{C,t}} = (1+\tau^f) e_t, \tag{5}$$

$$u_{C,t} = i_t \beta E_t \left[\frac{u_{C,t+1}}{\pi_{t+1}} \right]. \tag{6}$$

Interpretation: The domestic Euler equation is central for default amplification. If default risk raises expected future marginal utility, current consumption must fall unless monetary policy changes.

3.4 Firms and the New Keynesian Phillips curve

Final-goods firms aggregate differentiated intermediate varieties. Intermediate firms produce with labor:

$$y_{j,t} = z_t n_{j,t}. \tag{7}$$

Productivity depends on the aggregate productivity shock and credit standing:

$$z_t = z(\tilde{z}_t, \Delta_t). \tag{8}$$

During bad credit standing after default, productivity is lower.

Intermediate firms face Rotemberg pricing costs. The symmetric first-order condition gives a New Keynesian Phillips curve:

$$(1-\tau)\frac{w_t}{z_t} = \frac{\eta - 1}{\eta} + \frac{1}{\eta}\left[\phi(\pi_t - \bar{\pi})\pi_t - E_t \beta \frac{Y_{t+1} u_{C,t+1}}{Y_t u_{C,t}} \phi(\pi_{t+1} - \bar{\pi})\pi_{t+1} \right]. \tag{9}$$

Interpretation: Inflation today depends on current real marginal costs and expected future inflation weighted by marginal utility. This is the channel through which default risk changes current inflation.

3.5 Government debt, default, and spreads

The government issues long-term foreign-currency debt. If it issues ℓ_t and inherited debt is B_t, next-period debt is

$$B_{t+1} = (1-\delta)B_t + \ell_t. \tag{10}$$

Each unit of outstanding debt promises $r+\delta$ each period. The bond price satisfies the lenders' break-even condition:

$$q_t = \frac{1}{1+r} E_t \left[(1-D_{t+1})(r+\delta+(1-\delta)q_{t+1}) \right]. \tag{11}$$

The sovereign spread is

$$spread_t = (r+\delta)\frac{1}{q_t} - 1. \tag{12}$$

Interpretation: Spreads are endogenous. They rise when future default probabilities rise, which makes debt prices fall.

3.6 Government budget constraint

When the government has good credit standing and repays, the domestic-currency budget constraint is

$$T_t + \tau W_t N_t = \varepsilon_t \left[q_t(B_{t+1} - (1-\delta)B_t) - (r+\delta)B_t \right] + \tau^f P_t^f C_t^f. \tag{13}$$

In domestic-goods units this becomes

$$t_t + \tau w_t N_t = e_t \left[q_t(B_{t+1} - (1-\delta)B_t) - (r+\delta)B_t \right] + \tau^f e_t C_t^f. \tag{14}$$

Interpretation: Debt operations affect government transfers and therefore the private allocation. Exchange-rate movements also matter because debt is denominated in foreign currency.

3.7 Monetary policy rule

The baseline interest-rate rule is

$$i_t = \bar{i} \left(\frac{\pi_t}{\bar{\pi}} \right)^{\alpha_P} m_t, \tag{15}$$

where $\alpha_P > 1$ and m_t is a monetary policy shock.

The paper also studies:

- strict inflation targeting, $\pi_t = \bar{\pi}$;
- a rule that reacts directly to default risk;
- discretionary monetary policy.

Interpretation: The baseline rule resembles inflation-targeting central banks. The alternative rules are used to study whether targeting inflation alone is optimal once default risk is present.

3.8 Resource constraints and monetary wedge

The domestic goods resource constraint is

$$C(S) + X(S) + \frac{\phi}{2}(\pi - \bar{\pi})^2 Y(S) = Y(S), \tag{16}$$

where $Y(S) = z(\tilde{z}, \Delta)N(S)$.

The balance-of-payments condition is

$$e(S)C^f(S)(1 + \tau^f) - X(S) = t(S) + \tau w(S)N(S). \tag{17}$$

The paper defines the monetary wedge as

$$1 + \text{monetary wedge} = \frac{z(\tilde{z}, \Delta)}{w(S)} = -\frac{z(\tilde{z}, \Delta)u_C(S)}{u_N(S)}. \tag{18}$$

Interpretation: The wedge measures production inefficiency from sticky prices and monetary distortions. Production efficiency requires the marginal product of labor to equal the household marginal rate of substitution.

3.9 Recursive government problem

Let $V(s, \nu, B)$ be the government value including the default option:

$$V(s, \nu, B) = \max_{D \in \{0,1\}} \left\{(1 - D)W(s, B) + D[W^d(s) - \nu]\right\}. \tag{19}$$

The repayment value is

$$W(s, B) = \max_{B'} \left\{u(C, C^f, N) + \beta_g E V(s', \nu', B')\right\}. \tag{20}$$

The default value is

$$W^d(s) = u(C, C^f, N) + \beta_g E\left[\iota V(s', \nu', 0) + (1 - \iota)W^d(s')\right]. \tag{21}$$

The default cutoff satisfies

$$\hat{\nu}(s, B) = W^d(s) - W(s, B). \tag{22}$$

Default occurs when $\nu \leq \hat{\nu}(s, B)$.

Interpretation: Default is a threshold rule in the enforcement shock. More debt lowers repayment value and raises default probability.

3.10 Optimal borrowing condition

The optimality condition for borrowing is

$$u_{C^f}\left[q + \frac{dq}{dB'}(B' - (1 - \delta)B)\right](1 - \tau_m^X) - u_{C^f}\tau_m^C = \beta_g E(1 - D')u'_{C^f}(r + \delta + (1 - \delta)q')(1 - \tau_m^{X'}). \tag{23}$$

Without monetary frictions, the wedges are zero and the standard international Euler equation is

$$qu_{C^f} = \beta_g E\left[u'_{C^f}(r + \delta + (1 - \delta)q')\right]. \tag{24}$$

Interpretation: The monetary wedges are the key new objects. They act as additional borrowing costs and can discipline overborrowing caused by debt dilution and government impatience.

4 Theoretical characterizations

4.1 Default amplification

The simplified NKPC can be written as

$$(\pi - \bar{\pi})\pi = \frac{\eta - 1}{\phi}\left[-\frac{u_N}{z u_C} - 1\right] + \frac{\beta}{Y u_C} E\left[Y' u'_C (\pi' - \bar{\pi})\pi'\right]. \tag{25}$$

Default risk raises current inflation through three channels:

- default states have low productivity, so expected future inflation rises;
- default states have low consumption, so expected future marginal utility rises;
- the higher expected marginal utility makes future inflation more heavily weighted in current pricing decisions.

The domestic Euler equation is

$$u_C = i\beta E\left[\frac{u'_C}{\pi'}\right]. \tag{26}$$

When default risk raises expected future marginal utility, current consumption falls, which lowers output and increases the monetary wedge.

Interpretation: Default amplification is an expectations channel. Firms and households react today to the possibility that tomorrow's economy will be in default, low-consumption, high-inflation states.

4.2 Proposition 1: amplification

Proposition 1. Higher borrowing increases default risk, inflation, the nominal domestic rate, and the monetary wedge.

- Higher borrowing raises the probability of future default.
- Higher default risk raises expected inflation and expected marginal utility.
- The NKPC implies higher current inflation.
- The Euler equation implies lower current consumption.
- The interest-rate rule raises the nominal rate in response to higher inflation.
- The monetary wedge rises because output and consumption are inefficiently low.

Interpretation: Monetary policy dampens inflation but worsens the output/consumption wedge because tightening depresses demand.

4.3 Monetary discipline

Monetary discipline is the reverse interaction:

$$i \uparrow \quad \Rightarrow \quad \text{monetary wedge} \uparrow \quad \Rightarrow \quad \text{borrowing incentives} \downarrow \quad \Rightarrow \quad \text{default risk} \downarrow. \tag{27}$$

The government internalizes the fact that borrowing increases default risk and worsens monetary distortions. Tight policy therefore changes the government's borrowing Euler equation and discourages overborrowing.

Interpretation: Tight monetary policy is costly because it distorts domestic production, but that distortion can act as a commitment device against fiscal overborrowing.

4.4 Lemma 1: strict inflation targeting and overborrowing

Lemma 1. Strict inflation targeting can eliminate period-0 pricing distortions, but it yields higher default risk and lower welfare than the constrained-efficient allocation.

Interpretation: Removing monetary distortions is not enough. If fiscal overborrowing remains, the economy can still have inefficiently high spreads and default risk.

4.5 Lemma 2: default risk raises the monetary wedge

Lemma 2. Holding monetary policy fixed, higher default risk increases the monetary wedge when the wedge is positive.

Interpretation: Higher default risk raises expected marginal utility, which reduces current consumption. Lower consumption and real appreciation reduce demand for domestic goods and exports, raising the wedge.

4.6 Proposition 2: discipline

Proposition 2. If the policy rate is set above the strict-inflation-targeting rate, the monetary wedge is positive and default risk is lower than under strict inflation targeting.

The key government borrowing condition has an extra term because borrowing changes expected future marginal utility and hence the monetary wedge:

$$1 - h(\hat{\nu}(B'))B' - \frac{\kappa}{1 - F_{\nu}(\hat{\nu}(B'))} \frac{\partial E u'_C(B')}{\partial B'} \frac{u_C}{E u'_C(B')} = \beta_g(1 + r). \tag{28}$$

Interpretation: The multiplier κ on the domestic Euler equation makes borrowing more costly when borrowing worsens the wedge. This lowers default risk.

4.7 Proposition 3: default-risk monetary rule

Proposition 3. A monetary policy rule of the form

$$i = \bar{i}\,\Omega^{\alpha_D} \tag{29}$$

can implement constrained-efficient default risk and an arbitrarily small monetary wedge, for a positive default-risk response coefficient α_D.

Interpretation: A rule that reacts to default risk can replicate the fiscal discipline of tight monetary policy while keeping the production wedge close to zero. This is the paper's theoretical case for monetary rules that account for sovereign risk.

5 Quantitative analysis

5.1 Parameterization

- The model is calibrated and estimated to emerging-market inflation-targeting data.
- Assigned parameters include preference parameters, export demand elasticity, price-adjustment cost, productivity persistence, debt maturity, reentry probability, and monetary shock process.
- Moment-matched parameters include household discount factor, government discount factor, inflation target, Taylor-rule coefficient, productivity volatility, default-cost parameters, and enforcement shock magnitude.

- The inflation target implies about 4% annual inflation, in line with emerging-market targets.
- The rule coefficient $\alpha_P = 1.575$ lies within estimated ranges for the sample countries.

Interpretation: The calibration is designed to match both nominal moments and sovereign-risk moments. This is important because the paper's mechanisms link the two sets of variables.

5.2 Targeted moments

The model targets:

- mean CPI inflation;
- mean nominal domestic rate;
- mean sovereign spread;
- debt-to-output ratio;
- standard deviations of output, CPI inflation, spreads, and aggregate consumption;
- correlation between output and spreads.

Interpretation: Matching spreads, inflation, and output volatility simultaneously is a demanding exercise because default risk and monetary frictions jointly determine the moments.

5.3 Policy functions

- Figure IV plots policy rules as debt rises.
- Higher debt increases one-period-ahead default risk.
- Inflation, the monetary wedge, and the nominal policy rate all rise with debt.
- Consumption and output decline with debt because higher default risk tightens expectations and monetary policy.

Interpretation: The quantitative policy functions operationalize Proposition 1: higher debt raises default risk, which raises inflation and the wedge.

5.4 Impulse responses

- A contractionary monetary shock raises the domestic nominal rate.
- Inflation falls, consumption and output fall, and spreads decline.
- Spreads fall because higher rates discipline borrowing and reduce default probabilities.
- A negative productivity shock raises spreads and inflation and lowers output and consumption.
- Monetary policy responds to the inflationary pressure by increasing the nominal rate.

Interpretation: Monetary shocks show the disciplining channel. Productivity shocks show default amplification and the endogenous monetary response.

5.5 NK-Default versus NK-Reference

- The NK-Default model includes endogenous sovereign risk.
- The NK-Reference model eliminates default risk while keeping monetary frictions.
- Comparing the two isolates the contribution of sovereign risk to inflation, output, and consumption volatility.
- The paper finds that default risk accounts for about half of inflation volatility.

Interpretation: Default risk is not just a spread phenomenon. It is a major driver of nominal instability.

5.6 Temporary inflation event analysis

- The model reproduces the tent-shaped inflation event: inflation rises temporarily and then falls.
- Spreads rise together with inflation.
- Output falls during the event.
- Domestic rates rise as the monetary authority tightens.
- Without default risk, the model cannot generate the same inflation increase during spread events.
- Without the monetary response, inflation and spreads would be larger and more persistent.

Interpretation: The event analysis ties the mechanisms directly to the empirical patterns in emerging-market inflation events.

6 Robustness, alternative policies, and welfare

6.1 Robustness analysis

- The paper studies alternative monetary regimes, foreign-currency versus local-currency debt, shorter debt duration, alternative default costs, and domestic financial frictions.
- The baseline interaction between default risk and inflation remains robust.
- The disciplining role of monetary policy remains visible across environments.
- Removing productivity costs of default changes the quantitative moments but does not eliminate the mechanism.

Interpretation: The results are not an artifact of one specific calibration or one specific default-cost assumption.

6.2 Monetary discretion

- The paper compares simple rules with discretionary monetary policy.
- Discretion allows the monetary authority to condition policy on the state of the economy.
- However, discretion also faces the same trade-off: eliminating price distortions may worsen sovereign overborrowing.

Interpretation: The central issue is not whether policy is a rule or discretionary. The key issue is whether monetary policy internalizes sovereign default risk.

6.3 Welfare implications

- Welfare is reported as a consumption-equivalent measure relative to strict inflation targeting.
- The baseline monetary rule yields a small positive welfare gain over strict inflation targeting in the baseline economy.
- A rule that responds to default risk can deliver larger welfare gains and lower spreads.
- The welfare gains are especially visible in economies where monetary policy can lower default risk without creating excessive monetary distortions.

Interpretation: Strict inflation targeting is not generally optimal because it ignores the fiscal discipline role of monetary policy.

7 Tables and figures

7.1 Figure I and Table I: empirical motivation

Read:

- Figure I shows inflation, sovereign spreads, and domestic nominal rates during temporary inflation events.
- Inflation rises, spreads rise, and policy rates rise together.
- Table I reports the broader sample statistics for the eight emerging-market inflation targeters.
- Mean inflation is 3.9%, mean spreads are 2.0%, and spreads are positively correlated with inflation and domestic rates.

Economic message: The empirical motivation is the joint movement of nominal variables and sovereign risk. The model is designed to explain why spreads and inflation rise together.

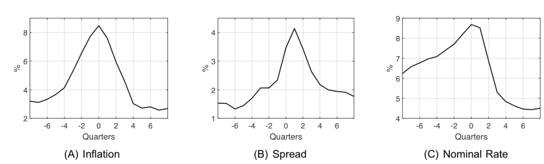


FIGURE I

Temporary Inflation Events

Figure I: temporary inflation events

TABLE I

EMERGING-MARKET INFLATION TARGETERS, KEY STATISTICS

	Mean		Std. dev. rel. output		Correlation with spread		
	Inflation	Spread	Inflation	Spread	Inflation	Domestic rate	Output
Brazil	5.6	2.8	0.6	0.3	0.5	0.6	−0.3
Chile	3.2	1.4	0.8	0.2	0.4	0.2	−0.6
Colombia	4.3	2.3	0.9	0.5	0.6	0.4	−0.3
Mexico	4.1	2.2	0.4	0.3	0.2	0.1	−0.5
Peru	2.9	2.0	0.5	0.3	0.5	0.1	0.0
Philippines	3.9	2.1	1.4	0.8	0.6	0.6	−0.5
Poland	2.1	1.1	0.9	0.4	0.4	0.2	−0.4
South Africa	5.0	2.2	1.2	0.5	0.5	0.1	−0.7
Mean	3.9	2.0	0.8	0.4	0.5	0.3	−0.5

Notes. Quarterly data, 2004Q1–2019Q4. Inflation is CPI four-quarter log difference. Spread is the JP Morgan EMBI+ Spread. Output is the four-quarter log difference of real GDP. See Online Appendix C.1 for the construction of our sample.

Table I: emerging-market statistics

7.2 Figure II and Figure III: event paths and monetary shocks

Read:

- Figure II plots average event paths for inflation, nominal rates, output, and spreads.
- Inflation and spreads follow tent-shaped paths, while output falls and recovers.
- Figure III projects spreads on monetary policy shocks.

- The empirical projection shows the disciplining relation: tighter monetary policy can lower spreads over time.

Economic message: The figures provide direct empirical discipline for the model's two mechanisms: amplification during inflation events and monetary discipline after monetary shocks.

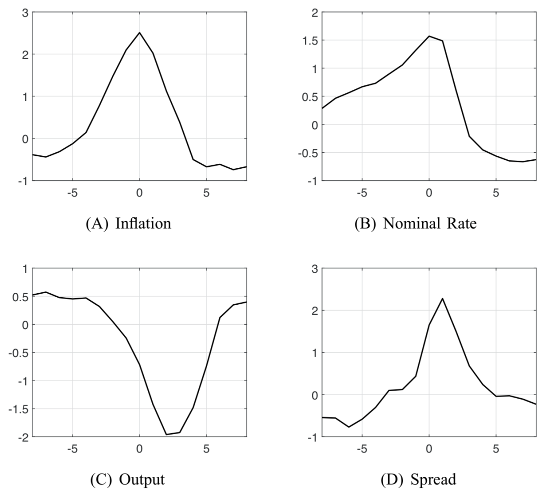


FIGURE II

Temporary Inflation Events

Figure II: event paths

FIGURE III

Projections of Spreads on Monetary Policy Shocks

Figure III: spreads and monetary shocks

7.3 Tables II and III: parameterization and targeted moments

Read:

- Table II reports assigned and moment-matched parameter values.
- The Taylor-rule coefficient is $\alpha_P = 1.575$ and the inflation target is $\bar{\pi} = 1.010$ quarterly.
- Debt duration is set to six years and market reentry probability implies a six-year average exclusion spell.
- Table III shows that the NK-Default model matches mean inflation, domestic nominal rates, spreads, debt, volatility, and output-spread correlation.

Economic message: The calibration ties the model to both inflation-targeting behavior and sovereign-risk dynamics.

TABLE II

PARAMETER VALUES

Assigned parameters		Parameters from moment matching	
Preferences	$\omega = 0.85, \theta = 0.73, \zeta = 0.33$	Discount factor	$\beta = 0.9947$
Varieties elasticity	$\eta = 6$	Interest rate rule	$\alpha_P = 1.575$
Export demand elasticity	$\rho = 5$	Inflation target	$\overline{\pi} = 1.010$
Price adjustment cost	$\varphi = 58$	Government discount factor	$\beta_g = 0.9775$
Productivity persistence	$\rho_z = 0.9$	Productivity volatility	$\sigma_z = 1.3\%$
Monetary shock	$\rho_m = 0.3, \sigma_m = 0.06\%$	Default costs	$\lambda_0 = -0.45, \lambda_1 = 0.5$
International rate	$r = 0.5\%$	Enforcement shock	$\varrho_D = 1.8e^{-4}$
Reentry probability	$\iota = 4.17\%$		
Debt duration	$\delta = 0.037$		

Table II: parameter values

TABLE III

TARGETED MOMENTS

	Data	NK-Default
Means		
CPI inflation	3.9	3.9
Nominal domestic rate	5.7	5.7
Spread	2.0	2.0
Debt to output	16	17
Standard deviations		
Output	2.3	2.2
CPI inflation	1.8	1.8
Spread	0.9	0.9
Consumption aggregate	2.4	2.6
Correlation		
(Output, spread)	−42	−53

cause none of the countries in our data sample outright defaulted during the sample period.[27] CPI inflation, nominal domestic rates, and spreads are reported annualized. In the model and data, CPI inflation is about 3.9%, spreads are 2%, nominal domestic rates are 5.7%, and debt is about 16% of output. The model also

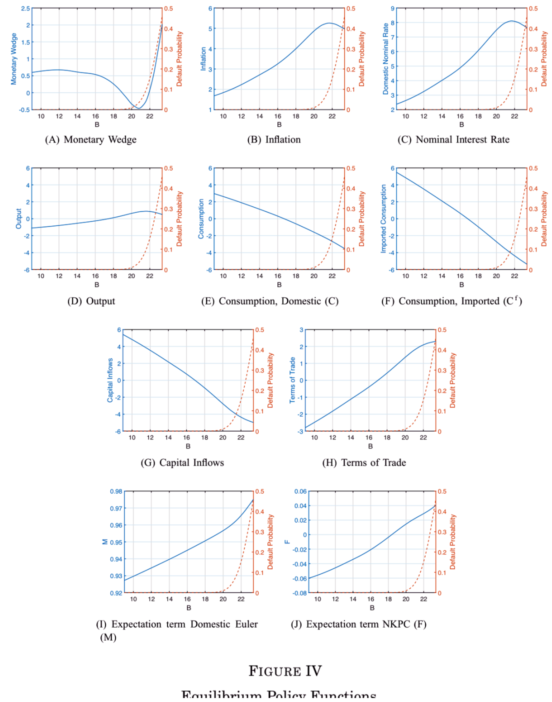
Table III: targeted moments

7.4 Figure IV: equilibrium policy functions

Read:

- Figure IV shows how inflation, policy rates, consumption, output, and wedges move with debt.
- Default risk rises with debt.
- Inflation and the policy rate rise as debt and default risk rise.
- Consumption and output fall as debt rises.

Economic message: The figure is the quantitative version of default amplification. High debt increases default risk, which raises inflation expectations and worsens monetary wedges.



7.5 Figures V and VI: impulse responses

Read:

- Figure V studies a contractionary monetary policy shock.
- Tight policy lowers inflation and spreads but depresses consumption and output.
- Figure VI studies a productivity shock.
- A negative productivity shock raises inflation, spreads, and policy rates while lowering output.

Economic message: Monetary shocks highlight discipline; productivity shocks highlight amplification.

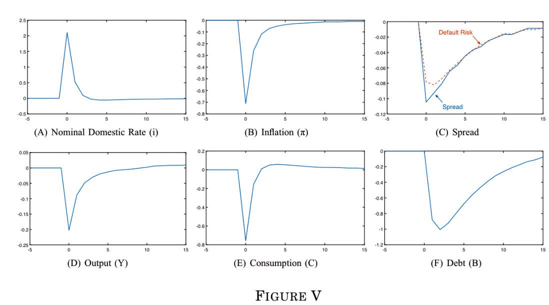


Figure V: monetary shock IRFs

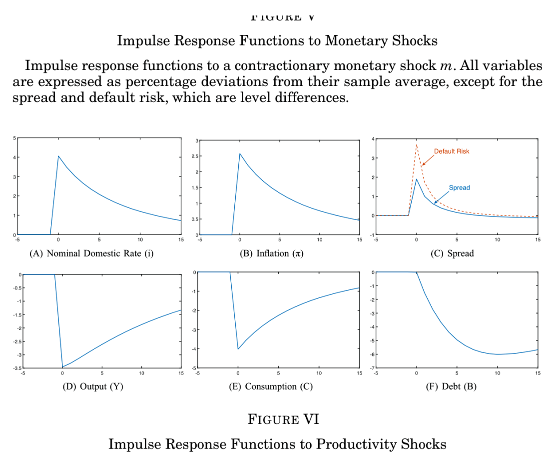


Figure VI: productivity shock IRFs

7.6 Table IV and Figure VII: model moments and event fit

Read:

- Table IV compares moments across the data, the baseline NK-Default model, and counterfactual models.
- Removing default risk reduces inflation volatility and changes the relation between spreads and macro variables.
- Figure VII compares temporary inflation events in the data and model.
- The baseline model reproduces the temporary rise in inflation, spreads, and policy rates together with an output decline.

Economic message: Default risk is quantitatively important for inflation volatility and for the model's ability to match event dynamics.

TABLE IV
COMPARISON OF MOMENTS ACROSS DATA AND MODELS

	Data	NK-Default	NK-Reference	NK-Default alternative rules: Strict inflation	NK-Default alternative rules: Inflation default
Mean					
CPI inflation	3.9	3.9	4.1	4.3	3.4
Nominal domestic rate	5.7	5.7	6.2	5.9	5.3
Spread	2.0	2.0	—	2.5	0.3
Debt	16	17	17	18	20
Standard deviation					
Output	2.3	2.2	3.3	2.3	2.3
CPI inflation	1.8	1.8	1.0	0.2	0.4
Spread	0.9	0.9	—	0.7	0.1
Consumption aggregate	2.4	2.6	1.6	3.3	3.7
Nominal domestic rate	1.9	3.1	1.6	2.3	3.8
Depreciation rate	8.6	1.9	1.6	1.0	1.1
Correlations					
(Spread, output)	−42	−53	—	−63	−59
(Spread, CPI inflation)	46	59	—	−13	59
(Spread, nom. dom. rate)	30	78	—	56	93
(Spread, depreciation rate)	37	52	—	−10	25
(CPI inf., output)	−22	−15	−90	7	−10
(CPI inf., nom. dom. rate)	59	90	90	−3	56
Autocorrelations (%)					
Output	84	69	94	71	74
CPI inflation	88	96	91	72	59
Spread	83	63	—	51	71
Nominal domestic rate	95	85	90	60	49

Table IV: moments across models

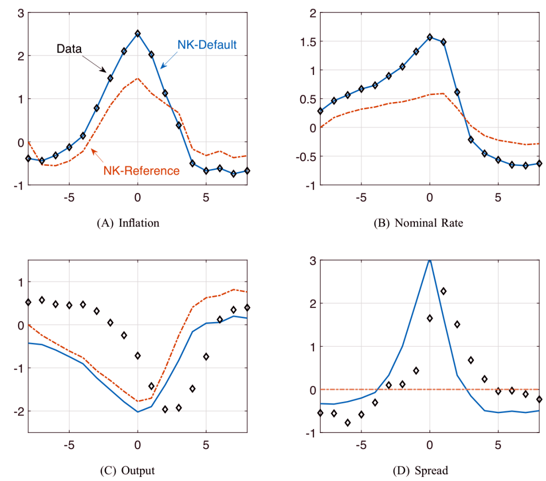


FIGURE VII
Inflation Event Analysis: Model Fit and the Role of Default Risk

Figure VII: event fit and default risk

7.7 Figure VIII and Table V: monetary policy role and robustness

Read:

- Figure VIII isolates the role of monetary policy in inflation events.
- Without appropriate tightening, inflation and spreads are higher and more persistent.
- Table V reports robustness across alternative economies and monetary discretion.
- The baseline mechanisms survive modifications to default costs, local-currency debt, shorter debt duration, and domestic financial frictions.

Economic message: The robustness results show that sovereign-risk-aware monetary policy remains valuable across model variants.

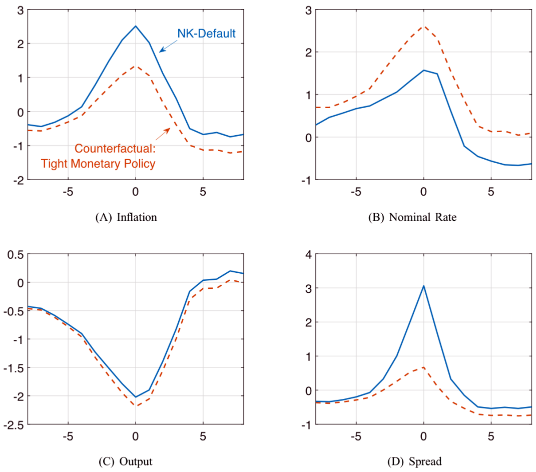


FIGURE VIII

Inflation Event Analysis: The Role of Monetary Policy

See the note to Figure VII. The solid blue line is the baseline model while the

Figure VIII: monetary policy in inflation events

TABLE V

ROBUSTNESS ANALYSIS AND MONETARY DISCRETION

		Local-currency debt		Alt. default costs		
	NK-Default	NK-Default	Discretion	Monetary	No prod.	Fin. frictions
Mean						
CPI inflation	3.9	4.1	9.3	3.9	4.2	3.8
Nominal domestic rate	5.7	6.0	10.8	5.8	6.1	5.6
Spread	2.0	0.6	1.5	1.6	0.8	2.0
Debt to output	17	23	23	19	18	18
Standard deviation						
Output	2.2	2.5	2.2	2.2	2.5	2.4
CPI inflation	1.8	1.7	0.4	1.9	1.7	1.8
Spread	0.9	0.3	0.4	0.7	0.1	1.0
Consumption aggregate	2.6	2.3	3.3	2.7	2.3	2.7
Nominal domestic rate	3.1	2.8	2.2	3.2	2.8	3.1
Depreciation rate	1.9	1.8	0.8	1.9	1.9	2.0
Correlations						
(Spread, output)	−53	−42	−62	−56	−3	−50
(Spread, CPI inflation)	59	71	9	56	16	61
(Spread, nom. dom. rate)	78	90	65	77	15	77
(Spread, depreciation rate)	52	60	9	49	7	51
(CPI inf., output)	−15	−3	−17	−17	−1	−12
(CPI inf., nom. dom. rate)	90	86	−10	88	88	87
Autocorrelations (%)						
Output	69	67	72	69	69	68
CPI inflation	96	95	85	96	98	96
Spread	63	71	48	61	85	65
Nominal domestic rate	85	83	55	84	89	85

Table V: robustness and discretion

7.8 Figure IX, Table VI, and Figure X: welfare

Read:

- Figure IX compares welfare across monetary policy regimes.
- Table VI reports welfare, mean spreads, and mean inflation across economies and rules.
- Figure X relates welfare gains to reductions in spreads across economies.
- Rules that respond to default risk can generate larger welfare gains than strict inflation targeting.

Economic message: Welfare is improved when monetary policy lowers default risk without generating excessive price distortions. This is why the flexible-price allocation is not automatically optimal.

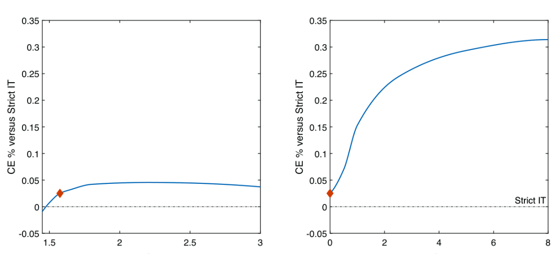


FIGURE IX

Welfare Comparison Across Monetary Policy Regimes

Welfare gains of the monetary rule (31) expressed in consumption-equivalent

Figure IX: welfare comparison

TABLE VI

WELFARE, SPREADS, AND INFLATION ACROSS NK-DEFAULT ECONOMIES AND MONETARY POLICY RULES

	Welfare (%)	Spread	Inflation
Baseline economy			
Strict inflation target	—	2.5	4.3
Baseline monetary rule	0.025	2.0	3.9
Less responsive monetary rule	−0.009	2.1	3.8
Inflation–default risk rule	0.314	0.3	3.4
Local-currency debt			
Strict inflation target	0.081	1.9	4.3
Baseline monetary rule	0.253	0.6	4.1
Inflation–default risk rule	0.354	0.1	4.4
Discretion	−0.195	1.5	9.3
Financial frictions			
Strict inflation target	0.020	2.3	4.3
Baseline monetary rule	0.025	2.0	3.8
Inflation–default risk rule	0.342	0.2	5.4
Monetary costs			
Baseline monetary rule pre-default	0.056	1.6	3.9
No productivity costs			
Baseline monetary rule	0.326	0.8	4.2
Shorter duration debt			
Strict inflation target	0.080	1.2	4.2
Baseline monetary rule	0.084	0.7	4.0

Notes: This table reports welfare, mean spread, and mean CPI inflation across economies and monetary policy rules. Welfare is consumption equivalence evaluated for zero debt and the mean level of shocks. Welfare is reported relative to the baseline economy under a strict inflation-targeting monetary policy.

Table VI: welfare, spreads, and inflation

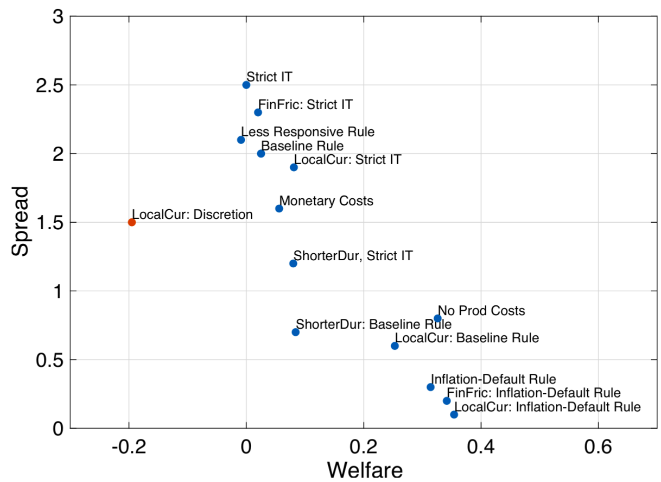


FIGURE X

Welfare and Spread Across Economies

8 Short summary

- The paper builds a New Keynesian sovereign-default model for emerging economies.
- Sovereign risk generates inflation pressure through expectations of high future inflation and low future consumption.
- Monetary policy affects sovereign risk because tight policy changes government borrowing incentives.
- The model explains temporary inflation events in emerging markets, in which inflation, spreads, and policy rates rise together and output falls.
- A baseline inflation-targeting rule can reduce spreads relative to strict inflation targeting.
- Strict inflation targeting is not necessarily optimal because it ignores fiscal overborrowing and default risk.
- A default-risk-responsive rule can improve welfare by disciplining borrowing while keeping inflation near target.
- Quantitatively, default risk accounts for a substantial part of inflation volatility.
- The paper reframes monetary policy in emerging markets as a joint nominal-stability and sovereign-risk-management problem.

9 Conclusion

9.1 Core conclusion

Arellano, Bai, and Mihalache show that sovereign risk and monetary policy interact in ways that standard New Keynesian models miss. Default risk raises inflation through expectations, and monetary policy can reduce default risk by disciplining government borrowing. Therefore, strict stabilization of inflation is not automatically welfare maximizing.

9.2 Economic intuition

Default risk makes firms and households expect bad future states: low consumption, lower productivity, and high inflation. These expectations raise current prices and depress current consumption. Monetary tightening reduces inflation and, more importantly, makes additional government borrowing more costly because borrowing worsens monetary distortions. The government internalizes these costs and borrows less.

9.3 Takeaway

For emerging-market central banks, sovereign spreads are not just fiscal variables. They affect inflation, output, and the optimal design of monetary policy. A policy framework that accounts for default risk can improve welfare relative to strict inflation targeting.